ICOSSAR 2021
The 13th International Conference on Structural Safety and Reliability (ICOSSAR 2021), June 21-25, 2021, Shanghai, P.R. China
J. Li, Pol D. Spanos, J.B. Chen & Y.B. Peng (Eds)

Modelling spatial correlation in earthquake-induced damage and its impact on regional loss estimation

Michele Nguyen[a] , Sabine Loos [b] and David Lallemant[c]

[a]Asian School of the Environment, Nanyang Technological University, Singapore, E-mail: michele.nguyen@ntu.edu.sg
[b]School of Engineering, Stanford University, Stanford, California, U.S.A., Email:sloos@stanford.edu
[c]Earth Observatory of Singapore, Nanyang Technological University, Singapore, Email: dlallemant@ntu.edu.sg

ABSTRACT: While correlation in ground motion intensity and its effect on earthquake damage has been well-documented, there is a widely held yet insufficiently examined view, that there are other sources of spatial correlation in damage not accounted for. To provide further evidence and illustrate impacts on regional losses, we develop a spatial ordinal damage model. This involves adding Gaussian random fields at a latent level and using data from multiple building types to separate the sources of spatial correlation in the absence of sensor data. Unlike previous approaches which require two fitting steps, this novel model is fit once using maximum likelihood estimation. As illustrated using data from the 2010 Haiti earthquake, the modelled spatial correlation leads to greater regionalization of the latent variable mean surface and shifts in damage state distributions. This highlights the increased probabilities of extreme regional loss due to the additional factors for spatial correlation in damage.

KEYWORDS: Spatial correlation; random fields; aggregate seismic loss.

1 INTRODUCTION

Regional loss estimation is an important tool to investigate the potential impact of a hazard such as an earthquake or a flood on spatially distributed assets. This concerns multiple stakeholders including municipal governments aiming to maintain economic activity after an event, insurance companies covering large spatially distributed portfolios, and lifeline systems engineers dealing with the impact on network infrastructure and services. In all these cases, it is the simultaneous damage to groups of assets that is of primary interest. For this reason, capturing the various sources of correlation in damages is critical to properly predict potential losses (Weatherill et al. 2015).

In the context of earthquakes, ground motion prediction equations (GMPEs) are traditionally used to describe attenuation relationships for single locations and predict ground motion intensity measures (IMs) such as peak ground acceleration (PGA). If these perfectly model the spatial characteristics of IM, one would expect to see no spatial pattern in its residuals when compared to sensor data. However, in various studies, intra-event, spatial correlation has been observed and variograms are typically used to model these (Jayaram & Baker 2009, Park et al. 2007, Silva 2019, Zafarani et al. 2021). As suggested by Park et al. (2007), this spatial correlation could be due to common wave paths and similar distances from asperities. Since increased spatial dependence of IMs increases the probability of the simultaneous occurrence of large intensities across a ground motion field, this leads to increased probabilities of extreme losses.

While correlation in earthquake damage due to that in the ground motion intensity measure is well-documented in literature, there is a widely held yet insufficiently examined view that there are other sources of spatial correlation left unaccounted for. Shome et al. (2012) begin to investigate this by modelling correlation in the differences between observed losses and damage ratio predictions from 4-parameter Beta distributions fitted for various spectral acceleration ranges. Such residual spatial correlation in damage can be attributed to local soil conditions, similarities in construction age and material, a domino effect between buildings

as well as the fact that there are other important measures of shaking such as duration and frequency content which have been omitted in the modelling process (Du et al. 2020).

In this paper, we provide further evidence of spatial correlation in earthquake damage and show the impact this has when fitting fragility curves to empirical data and estimating regional losses. As well as having implications for multiple stakeholders, our work presents several methodological advances. Instead of a two-step approach of accounting for correlation in IM or fitting 4-parameter Beta distributions to loss ratios before modelling spatial correlation in damage (Park et al. 2007, Jayaram & Baker 2009, Silva 2019), we develop a novel damage model which can be fit in one step. This can be seen as a spatial extension of the ordinal regression or cumulative link model. This has been used to relate building damage to an IM such as PGA while borrowing information across damage states and fitting fragility curves for all states concurrently (Lallemant et al. 2015). By modelling spatial correlation in damage at a latent variable level, we can relate fitted variograms to the covariances of Gaussian spatial random fields and simulate from our fitted model in a statistically consistent way. By using damage data for two building types in the absence of sensor data, we can separate the shared effects of spatial correlation in the IM from the spatial correlation in damage itself.

We begin in Section 2 by introducing the use of variograms to measure spatial dependence and providing a quick illustration of spatial correlation in damage with spatially aggregated mean damage from the 2010 Haiti earthquake. This motivates our development of a more formal model to address this. In Section 3, we recap the key components of a standard ordinal regression to work towards the proposed spatial ordinal model. With the models outlined, we demonstrate the effects of spatial correlation in damage in Section 4 by fitting the proposed damage model to the Haiti dataset and comparing estimated regional portfolio losses to those obtained from its submodels: a non-spatial ordinal regression and a spatial ordinal model which only accounts for correlation in the IM and not other factors. Since the majority of the buildings in Haiti are short and stiff, we use PGA as the IM in our models to account for the high-frequency motion which they are sensitive to. Note that other IMs such as spectral acceleration could be used in place of PGA in the model for other study regions where tall or mid-height buildings are dominant and shaking at longer periods plays an important role.

Even though it is presented in the context of earthquakes, the spatial ordinal model can be adapted for other hazards and applications. For example, damage from floods is often modelled using water height in a univariate way; however, the flow velocity and duration of flooding may be important and spatially correlated as well (Wagenaar 2012). In the context of tropical cyclones, Zeng et al. (2020) show via model simulations how spatial correlation in both wind speeds and structural vulnerability affects the collective damage to spatially distributed building portfolios. Across the different hazards, the unifying idea is that there is a common shock but the response curve is different for different exposure categories (for example, of buildings) and there are unknown spatially correlated factors. In line with the growing recognition of the need to develop tools to address the complexities of spatial risk analysis (Argyri et al. 2019), a R package containing the functions we used to fit and evaluate the model is under development and plans are in line to make it publicly available with example analyses.

2 MEASURING SPATIAL CORRELATION

2.1 *In seismic ground motion intensity*

For any given event, the IM at site j can be modelled using the mean prediction from a GMPE, IM^{GMPE}, and a site-to-site variability or residual term u_j (Abrahamson & Silva 2008, Boore & Atkinson 2008):

$$\log\left(IM_j^{actual}\right) = \log\left(IM_j^{GMPE}\right) + u_j.$$

Using ground-motion recordings from earthquakes at spatially distributed sites, we can calculate the residual terms $u_j = u(\boldsymbol{x}_j)$ and measure their spatial dependence through a variogram:

$$\gamma(h) = Var(u(\boldsymbol{x}+\boldsymbol{h}) - u(\boldsymbol{x})) = E\left(\left(u(\boldsymbol{x}+\boldsymbol{h}) - u(\boldsymbol{x})\right)^2\right)$$

where $\mathbf{x} \in R^d$ is a d-dimensional spatial location and $\boldsymbol{h} \in R^d$ is a displacement vector. The variogram describes the variance of the difference between two variables, $u(\boldsymbol{x}+\boldsymbol{h})$ and $u(\boldsymbol{x})$ as a

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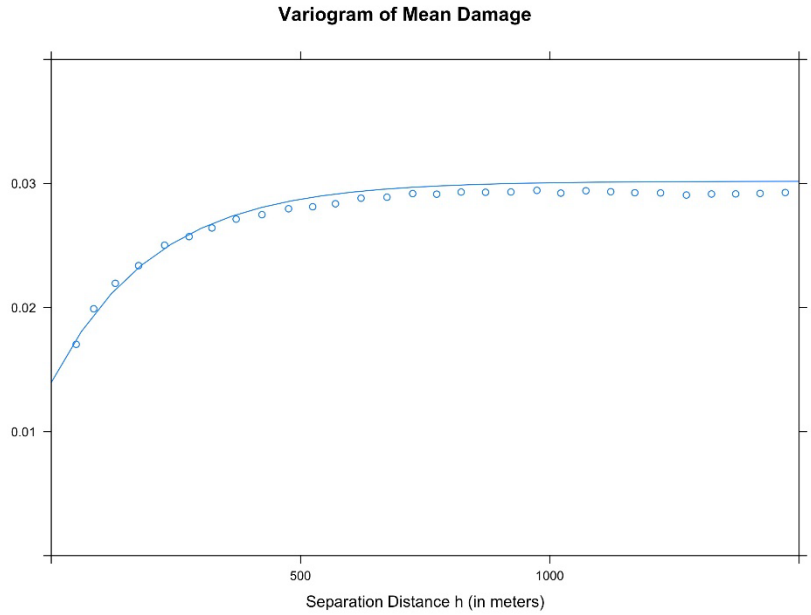


Figure 1. Estimated variogram of mean damage from post-earthquake data from the 2010 Haiti earthquake. The blue curve denotes the fitted exponential model.

function of their separation distance, $h = |\boldsymbol{h}|$ (assuming intrinsic stationarity), and can be estimated from empirical data as (Chilès & Delfiner 1999):

$$\hat{\gamma}(h) = \frac{1}{N(h)} \sum_{|\boldsymbol{x}_\beta - \boldsymbol{x}_\alpha| = h} \left(u(\boldsymbol{x}_\beta) - u(\boldsymbol{x}_\alpha)\right)^2, \qquad (1)$$

where $\boldsymbol{x}_\beta$ and $\boldsymbol{x}_\alpha$ denote sample locations, and $N(h)$ is the number of pairs in the dataset with distance h apart.

2.2 *In spatially aggregated mean damage from the 2010 Haiti earthquake*

In the same way that the variogram has been used to measure spatial dependence in IM, it can be used to support the presence of spatial correlation in damage. To this end, we use damage data from the very extensive field surveys conducted and led by the Haitian Ministry of Public Works in the months following the 2010 earthquake in Haiti. The data consists of over 400,000 data-points corresponding to each building assessed in the Port-au-Prince area and several nearby cities. The level of damage was determined using the ATC-20 methodology (Graf & Seligson 2011) following the damage scale used in ATC-13 (Applied Technology Council 2012). It classifies damage into seven damage states: no damage, insignificant damage, minor damage, moderate damage, extensive damage, extreme damage and complete collapse. Over 300 engineers were trained to conduct post-earthquake damage assessment by leading earthquake engineering experts, and quality assurance was provided by United Nations and third-party structural engineering experts.

For a quick illustration of spatial correlation in damage, we work with spatially aggregated mean damage. The individual building points were translated into spatial aggregates at a resolution of 50m x 50m and mean damage is calculated as a normalized average for all building damage within each 50m x 50m grid square:

$$\overline{\mu_{DS_j}} = \frac{1}{6n_j} \sum_{s=1}^{n_j} (k_s - 1), \qquad (2)$$

where $\overline{\mu_{DS_j}}$ is the observed (sample) mean, k_s is the damage state of buildings (range from 1-7) and n_j is the number of buildings in grid location j. The result is therefore a value of mean damage ranging from zero to one.

The variogram of Figure 1 demonstrates a spatial correlation in damages within a separation distance range of 500m. Since the assumption of

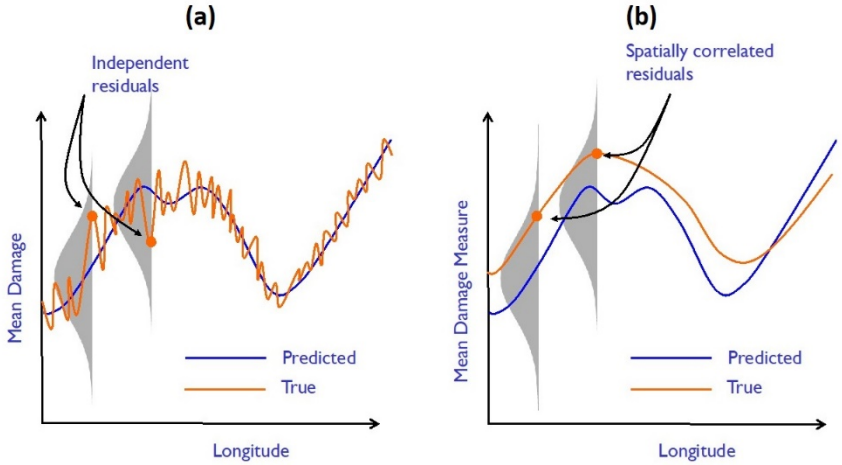


Figure 2. Simulation of true mean damage assuming: (a) independent; (b) spatially correlated residuals.

near-constant ground-motion intensity measure is reasonable within this distance (Jayaram & Baker 2009, Silva 2019, Park et al. 2007, Zafarani et al. 2021), this suggests that there is spatial correlation in damage beyond that explained by ground motion intensity.

Figure 2 illustrates the importance of modelling spatial correlation in damage. While a fragility model allows us to predict mean damage according to ground motion intensity, we typically observe differences between the predictions (represented by the blue curves) and actual observations (orange curves). These residuals are typically modelled as independent, as illustrated in the left plot. This assumes that they would have little to no impact on the predicted loss for a relatively large portfolio. If however, these residuals are spatially correlated, as depicted in the right plot and suggested from the Haiti case study, the mean damage could be systematically higher (or lower) than predicted over a large area.

2.3 *Towards a formal model*

Since the spatial dependence in damage is important to regional portfolio loss estimation, we work towards a more formal damage model which properly accounts for it. The spatial ordinal model to be introduced in the next Section aims to improve upon the quick illustration in Section 2.2 by:

1. treating damage states as ordinal instead of numerical;
2. making a more explicit distinction between the spatial correlation in IM and a spatial correlation in damage apart from that in IM;
3. using point locations instead of grid squares for better inference on spatial ranges;
4. and using latent spatial fields which have covariance structures related to variogram models to adhere to the Gaussian distribution assumption commonly made for simulation.

3 DAMAGE SPATIAL CORRELATION MODEL

3.1 *Non-spatial ordinal model*

Damage data are usually generated by classifying physical damage according to an ordinal damage scale, such as the EMS-98 damage scale where Grade 0 refers to no damage and Grade 5 refers to collapse. While the conventional method of fitting lognormal fragility curves treats the damage states as nominal (distinct and unordered), ordinal regression retains the order of the states and borrows information across them to fit all the relevant fragility curves simultaneously (Lallemant et al. 2015, Nguyen & Lallemant 2021). In addition to improving sensitivity to the overall structure of the damage data, using an ordinal model solves an issue commonly faced by the conventional nominal approach: crossing fragility curves which erroneously implies negative damage state probabilities (Applied Technology Council 2012). Since more data is used to estimate each curve and the model is more parsimonious, i.e. able to explain the data with less parameters, the curves obtained from the ordinal model also have lower associated

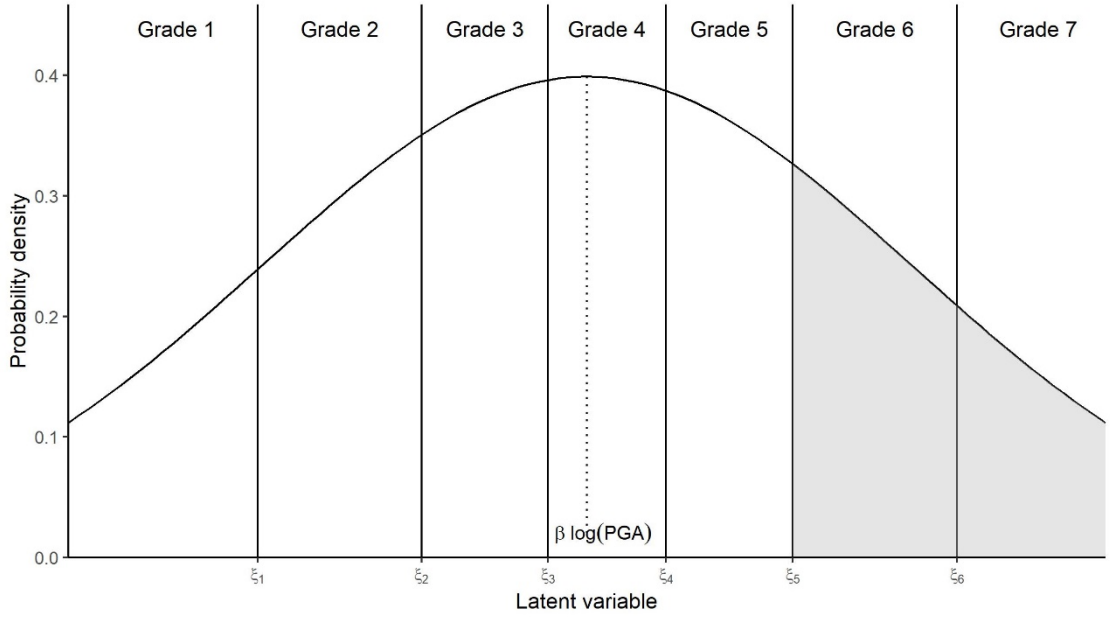


Figure 3. Plot of the latent variable distribution in a non-spatial ordinal model where the mean of the normal distribution depends on the $\log(IM)$ value. Here, the intensity measure (IM) is chosen to be peak ground acceleration (PGA). The probability of exceedance for a damage state is given by the area under the curve from the associated cut-off value which is denoted by a vertical line. The shaded area depicts this for the exceedance of State 5.

uncertainty and are less prone to overfitting.

Suppose we have K different damage states where $K \in \mathbb{N}$. In an ordinal model construction, we assume that there is a normally distributed latent variable $Z_j \sim N(\beta \log(IM_j), 1)$ for a building at location j. This means that its expected value depends on the IM at that location and $\beta \in R$ describes how responsive the building is to it. The building is said to be in damage state D_j if Z_j falls in the corresponding bin demarcated by the following cut-off points:

$$\xi_0 = -\infty < \xi_1 < \cdots < \xi_K = \infty. \quad (3)$$

This set-up is illustrated in Figure 3. Here, the breakpoints or cut-off values are represented by the bold vertical lines and the probability of having or exceeding a certain damage state can be calculated by the area under the normal density curve. For example, the probability of exceeding State 5 is given by the shaded area in Figure 3. Following the same argument, for $1 \le k \le K$:

$$P(D_j \le k) = P(Z_j \le \xi_k) = \Phi(\xi_k - \beta \log(IM_j)), \quad (4)$$

where Φ is the standard normal cumulative distribution function. This corresponds to the log-normal fragility curves which are also commonly used for single damage states.

Maximum likelihood estimation (MLE) can be used to obtain model parameter estimates. Based on our assumptions, the likelihood of our data takes the following form:

$$\mathcal{L}(\boldsymbol{\theta}) = P(D_1 = d_1, \dots, D_J = d_J) = \prod_{j=1}^{J} \left(P\left(Z_j \le \xi_{d_j}\right) - P\left(Z_j \le \xi_{d_j - 1}\right) \right) = \prod_{j=1}^{J} \left(\Phi\left(\xi_{d_j} - \beta \log(IM_j)\right) - \Phi\left(\xi_{d_j - 1} - \beta \log(IM_j)\right) \right) \quad (5)$$

Here, we have denoted the vector of model parameters to be estimated as $\boldsymbol{\theta} = \{\xi_1, \dots, \xi_{K-1}, \beta\}$. These are related to the more commonly used lognormal cumulative distribution curve parameters, namely the logarithmic standard deviation of the normal distribution and its median value, via $\tilde{\beta} = 1/\beta$ and $m_k = \exp(\xi_k/\beta)$, $1 \le k \le K$ respectively.

3.2 *IM-spatial ordinal model*

The non-spatial ordinal model relates damage state probabilities to the underlying IM values.

However, in practice, these are often simulated rather than observed values since we do not have sensor readings at every point of the study area. As mentioned by Park et al. (2007), there is a need to account for the residual spatial correlation in IM before estimating the damage resulting from it because it leads to higher probabilities of extreme regional portfolio losses. Although we can follow their proposed two-step strategy of simulating IM with a fitted variogram before fitting the damage model, the method is only viable when we have sufficient sensor readings spread out over the study area since we need these to properly estimate the variogram parameters. Since sensor readings might be unavailable, as in our case study, and spatial correlation in IM does not account for that in other shaking metrics, we propose to use building damage information on multiple building types to help us separate shared, ground motion spatial effects from effects which are associated with different building types.

Let $D_j^{(1)}$ and $D_j^{(2)}$ denote the damage states of building category 1 and category 2 at location j respectively. Then, instead of modelling them separately with two ordinal regressions, we can model them jointly as follows:

$$P\left(D_j^{(1)} \le k \middle| u_j^S\right) = P\left(Z_j^{(1)} \le \xi_k^{(1)} \middle| u_j^S\right),$$
$$\text{and } P\left(D_j^{(2)} \le k \middle| u_j^S\right) = P\left(Z_j^{(2)} \le \xi_k^{(2)} \middle| u_j^S\right), \quad (6)$$

where $\mathbf{Z^{(1)}} = \beta_1 (\log(\boldsymbol{IM})) + \boldsymbol{u}^S) + \boldsymbol{e}^{(1)}$ and $\mathbf{Z^{(2)}} = \beta_2 (\log(\boldsymbol{IM})) + \boldsymbol{u}^S) + \boldsymbol{e}^{(2)}$ are the latent variables corresponding to building category 1 and 2 respectively. These are linearly related to $\log(IM)$ just as in a traditional ordinal model.

The difference between the traditional model and an IM-spatial ordinal model is the addition of three terms: $\boldsymbol{u}^S$, $\boldsymbol{e}^{(1)}$ and $\boldsymbol{e}^{(2)}$. The first term $\boldsymbol{u}^S \sim GP(0, \Sigma)$ denotes a common spatial field which is modelled using a Gaussian process (GP) with a spatial e.g. Matérn covariance matrix, Σ. This captures the residual spatial correlation in $\log(\boldsymbol{IM})$ which is present because of differences between the modelled and actual IM values across the study area. To account for micro-scale variability in the damage data which could be due to measurement error, we include independent, identically distributed (i.i.d.) random error terms $e_j^{(1)} \sim N(0, \tau_1^2)$ and $e_j^{(2)} \sim N(0, \tau_2^2)$.

Similar to the previous section, our model assumptions mean that:

$$\begin{aligned} &P\left(D_j^{(1)} = d_j \middle| u_j^S\right) \\ &= \Phi\left(\frac{\xi_{d_j}^{(1)} - \beta_1\left(\log(IM_j) + u_j^S\right)}{\tau_1}\right) \\ &\quad - \Phi\left(\frac{\xi_{d_j-1}^{(1)} - \beta_1\left(\log(IM_j) + u_j^S\right)}{\tau_1}\right), \\ &P\left(D_j^{(2)} = d_j \middle| u_j^S\right) \\ &= \Phi\left(\frac{\xi_{d_j}^{(2)} - \beta_2\left(\log(IM_j) + u_j^S\right)}{\tau_2}\right) \\ &\quad - \Phi\left(\frac{\xi_{d_j-1}^{(2)} - \beta_2\left(\log(IM_j) + u_j^S\right)}{\tau_2}\right). \end{aligned} \quad (7)$$

Assuming that the coding of damage states is independent across damage states given the latent variable, we have:

$$\begin{aligned} \mathcal{L}(\theta) &= P\left(D_1 = d_1, \ldots, D_J = d_J\right) \\ &= \int \prod_{j=1}^{J} P\left(D_j = d_j \middle| u_j^S\right) P(\boldsymbol{u}^S) \mathrm{d}\boldsymbol{u}^S. \end{aligned}$$

33 *Damage-spatial ordinal model*

To account for spatial correlation in damage that is not fully explained by that in ground motion, we can extend the model in the previous section as follows:

$$P\left(D_j^{(1)} \le k \middle| u_j^{(1)}, u_j^S\right) = P\left(Z_j^{(1)} \le \xi_k^{(1)} \middle| u_j^{(1)}, u_j^S\right),$$
$$P\left(D_j^{(2)} \le k \middle| u_j^{(2)}, u_j^S\right) = P\left(Z_j^{(2)} \le \xi_k^{(2)} \middle| u_j^{(2)}, u_j^S\right). \quad (8)$$

Here, $\boldsymbol{Z}^{(1)} = \beta_1(\log(\boldsymbol{IM}) + \boldsymbol{u}^S) + \boldsymbol{u}^{(1)} + \boldsymbol{e}^{(1)}$ and $\boldsymbol{Z}^{(2)} = \beta_2(\log(\boldsymbol{IM}) + \boldsymbol{u}^S) + \boldsymbol{u}^{(2)} + \boldsymbol{e}^{(2)}$ where $\boldsymbol{u}^{(1)} \sim GP(0, \Sigma_1)$ and $\boldsymbol{u}^{(2)} \sim GP(\mathbf{0}, \Sigma_2)$ are building category specific spatial fields. Just as how $\boldsymbol{u}^{(S)}$ represents the long-range spatial correlation in IM, these terms represent the shorter-range spatial correlation in damage which

are not explained by that in IM. For example, if there are standard building practices or a big developer developing many buildings in an area, this could influence the terms $\boldsymbol{u}^{(1)}$ and $\boldsymbol{u}^{(2)}$. Since different building categories may be affected by these factors to different degrees, we estimate a separate spatial term for each building categories.

Similar to the previous section, we have that:

$$P\left(D_j^{(1)} = d_j \middle| u_j^{(1)}, u_j^S\right) = \Phi\left(\frac{\xi_{d_j}^{(1)} - \left(\beta_1\left(\log\left(IM_j\right) + u_j^S\right) + u_j^{(1)}\right)}{\tau_1}\right) - \Phi\left(\frac{\xi_{d_j-1}^{(1)} - \left(\beta_1\left(\log\left(IM_j\right) + u_j^S\right) + u_j^{(1)}\right)}{\tau_1}\right),$$

$$P\left(D_j^{(2)} = d_j \middle| u_j^S\right) = \Phi\left(\frac{\xi_{d_j}^{(2)} - \left(\beta_2\left(\log\left(IM_j\right) + u_j^S\right) + u_j^{(2)}\right)}{\tau_2}\right) - \Phi\left(\frac{\xi_{d_j-1}^{(2)} - \left(\beta_2\left(\log\left(IM_j\right) + u_j^S\right) + u_j^{(2)}\right)}{\tau_2}\right). \quad (9)$$

The likelihood $\mathcal{L}(\theta)$ can be computed analogously.

4 CASE STUDY: HAITI 2010 EARTHQUAKE

4.1 *Fitted damage model*

To fit the spatial ordinal model to the 2010 Haiti earthquake damage data, we use a balanced subset of the data consisting of 98 randomly selected buildings for each damage state and building type. Building category 1 corresponds to buildings with unreinforced block walls while building category 2 corresponds to stone masonry buildings. All buildings were restricted to be single storied and on flat ground.

Figure 4 shows the fitted variograms for the shared and building category specific spatial fields. The estimated spatial range for the shared spatial field is about 20 km, which is comparable to literature estimates of 5-100km for different case studies (Jayaram & Baker 2009, Park et al. 2007, Silva 2019, Zafarani et al. 2021). The estimated spatial ranges for the building category specific spatial fields are both about 2km. This is different from the 500m obtained from empirical variogram of the spatially aggregated data and highlights the importance of model assumptions.

The ratio of nugget variance (y-intercept) to sill (where the curves flatten out) translates to the relative portion of variability that is random vs spatially correlated. The fitted variogram of the shared spatial field indicates that neighbouring buildings will experience near-identical shaking (by virtue of the near perfect spatial correlation over very short distance of the shared spatial field). However, from the fitted building category specific variograms, we see that even for a single building type, two neighbouring buildings will show random variability in damage (hence the non-zero nuggets) but still exhibit correlation (since they are neighbouring).

Figure 5(a) shows the mean surface of the latent variable estimated for building category 1 (unreinforced block walls) in Port-au-Prince, the capital of Haiti. Here, we have chosen two locations (Site 1 and 2) to illustrate the effect of the building category specific spatial field on the ordinal probability distribution. Although these two locations have similar PGA values, the building category spatial field leads to finer spatial patterns in the estimated latent variable. We can see this by comparing the surface in Figure 5(a) with that Figure 5(b) where the contribution of the building category specific field has been removed.

With reference to Figures 5(a) and 5(c), higher latent variable mean values near Site 1 lead to a shift in the latent variable probability density to the right. This in turn leads to higher exceedance probabilities for higher damage states as compared to that for locations near Site 2.

4.2 *Regional loss estimation*

To illustrate the importance of modelling spatial correlation in damage for regional loss estimation, we consider a building portfolio composing of 150 randomly chosen buildings of each building type within a 2km by 2km grid. We simulate annual losses according to the fitted damage-spatial ordinal model as well as its non-spatial and IM-spatial submodels. From (9), we can simulate damage state probabilities for a seismic event by simulating values for the spatially correlated error terms as well as the PGA. While the former two can be generated using the fitted normal distributions and GPs, we use the OpenQuake engine (Pagani et al. 2014) to generate PGA simulations corresponding to one million stochastic event sets

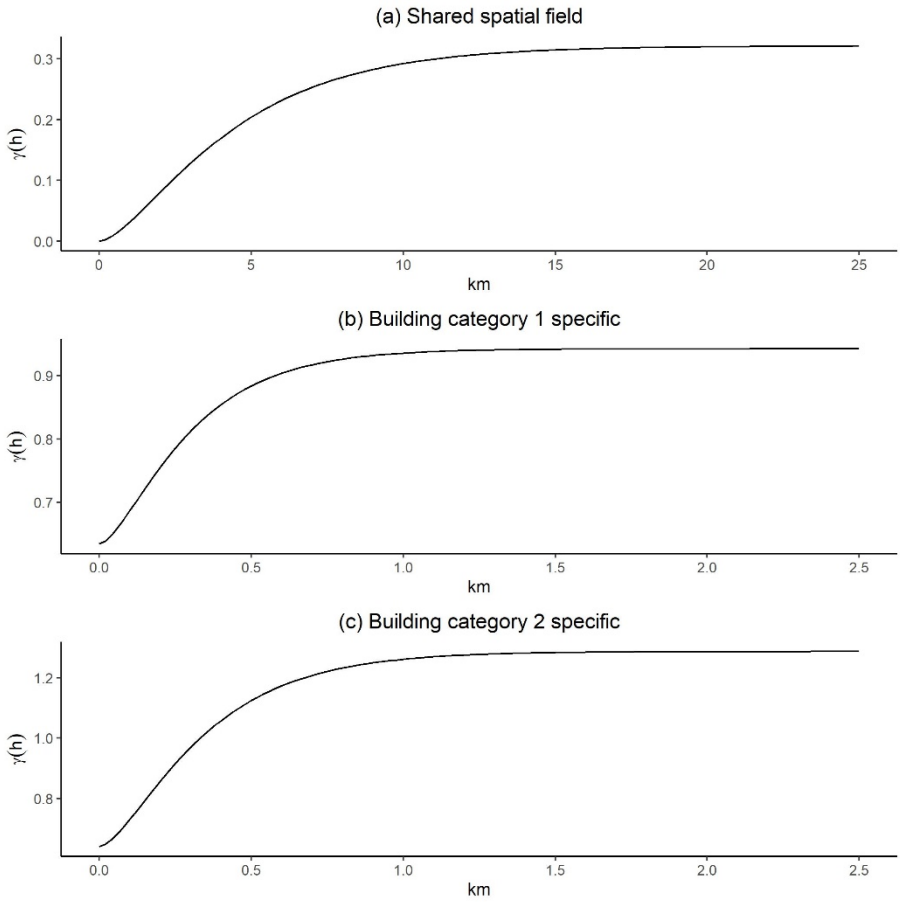


Figure 4. Fitted variograms for (a) the shared spatial field; (b)-(c) building category 1 and 2 specific spatial fields.

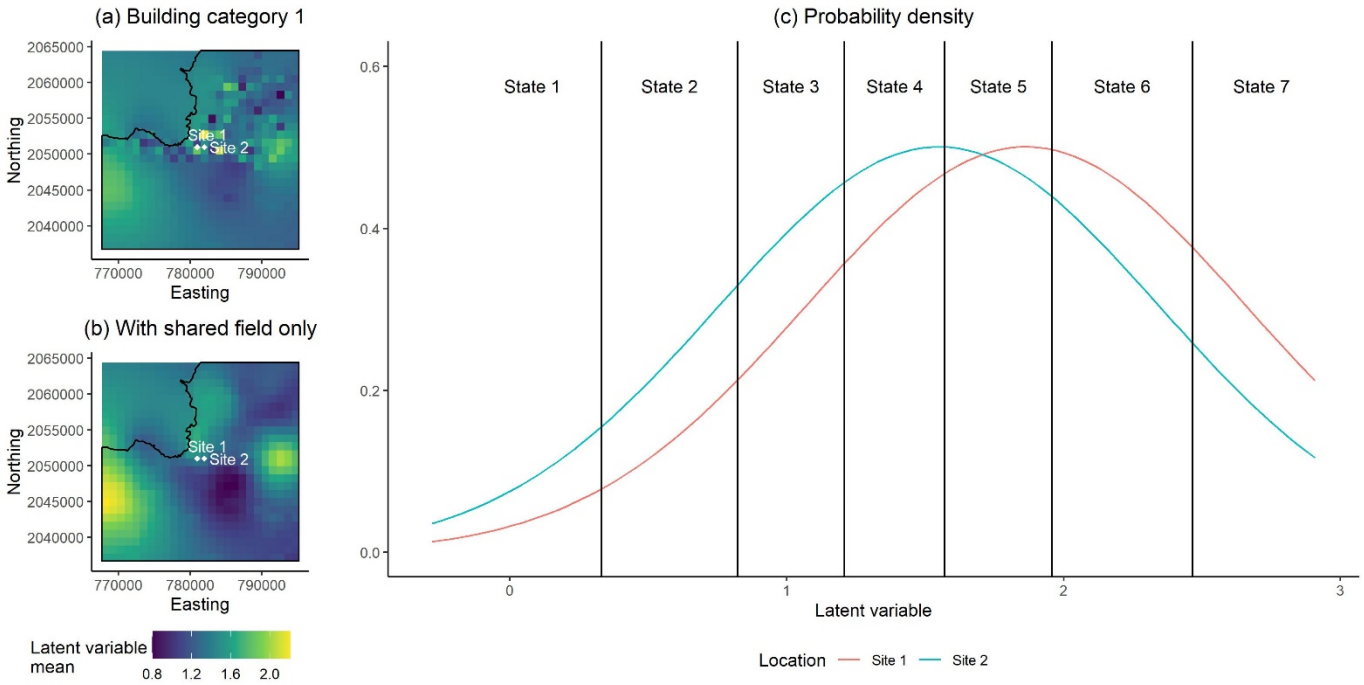


Figure 5. (a) Estimated latent variable mean surface for building category 1 in Port-au-Prince, Haiti; (b) the surface without the building category 1 specific field contribution; (c) the ordinal distributions at the two sites indicated according to the latent variable mean surface in Plot (a).

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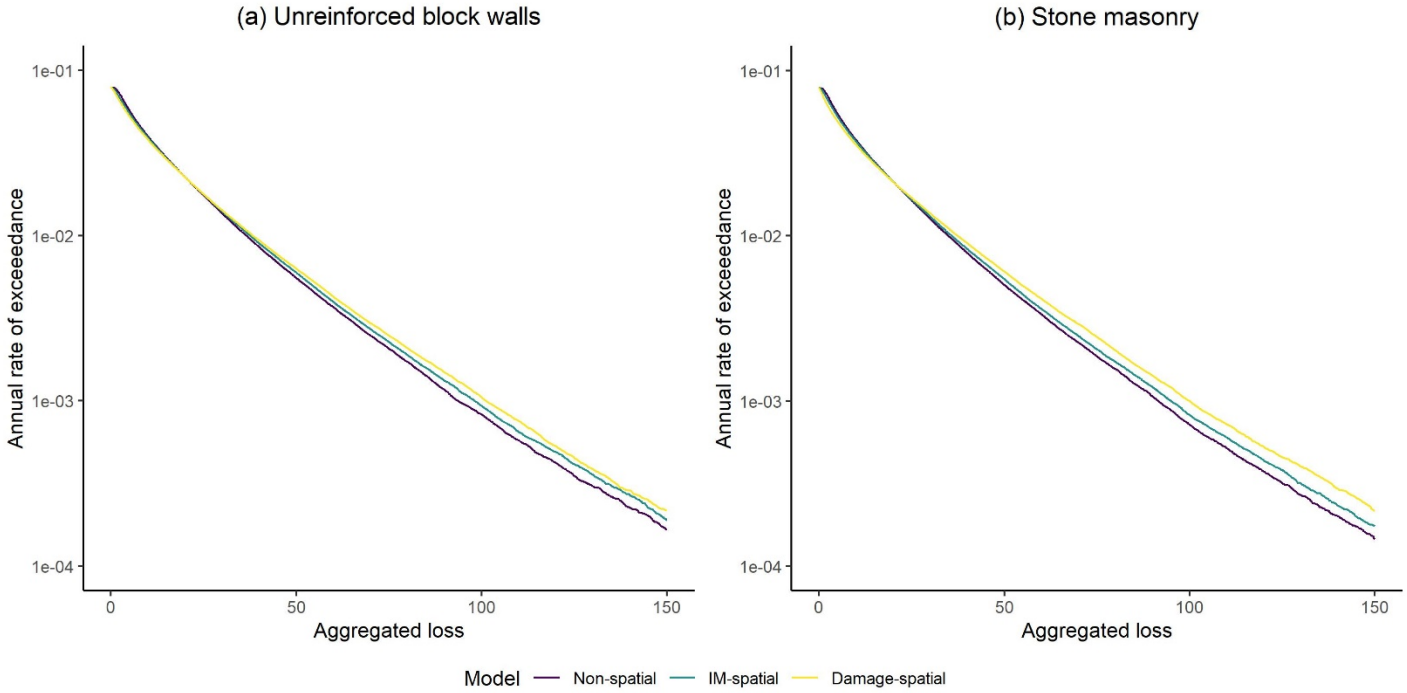


Figure 6. Loss exceedance curves for the portfolios of (a) 150 Category 1 (unreinforced block walls) and (b) 150 Category 2 (stone masonry) buildings within a 2km by 2km region in Port-au-Prince. The values on the x-axis are given in unit costs.

(SESs). Each SES corresponds to a duration of one year to enable us to compute annual losses. For each model, the estimated mean central damage factor (CDF) of a building after an event is computed as the weighted sum of the simulated damage state probabilities and their corresponding CDFs as defined by the damage scale. If we define the replacement cost of each building to be one unit, this is equal to the estimated mean replacement cost of the building. For each SES, the annual loss is calculated as the sum of the estimated mean replacement cost of the buildings in the portfolio due to events occurring within that simulated year.

Based on the annual losses per SES year, we plot the rates of exceedance in Figure 6. Accounting for correlation increases the probability of significant losses (as expected). For Building Category 2 (stone masonry), this increase due to spatial correlation in damage is on the same order as that produced by spatial correlation in IM. Hence it is important to account for this.

In Figure 6, the difference in the loss exceedance curves for the IM-spatial and Damage-spatial models is much smaller for Building Category 1 (unreinforced block walls) than Building Category 2. This is likely because the relative contribution of its spatial component $\boldsymbol{u}^{(1)}$ is smaller: the ratio of its partial sill (that is the difference between the sill and the nugget) to its sill is 0.327 while that for Building Category 2 is 0.503. When the nugget variance is large, random noise will drown out the spatial correlation effects. Since the partial sill to sill ratio for unreinforced masonry is lower than that for stone masonry, in engineering terms, this means that there is more variation in the responses for unreinforced masonry. This could be due to more variation in quality and could also be related to the fact that the category of unreinforced masonry is broader and comprised of buildings with large as well as small footprint while the category of stone masonry is likely to contain mainly standard residential homes.

The Akaike Information Criterion (AIC) values for the Non-spatial, IM-spatial and Damage-spatial models are calculated to be 5353.78, 5278.86 and 5136.21 respectively. Since the Damage-spatial model gives the lowest AIC value and fits the Haiti damage the best, this provides evidence for spatial correlation in damage beyond that explained by PGA.

5 CONCLUSION AND FURTHER WORK

Spatial correlation in earthquake building damage is not fully explained by that in the ground motion intensity measure (IM) and a static fragility curve

cannot fully describe the variation in damage. Using data from the 2010 Haiti earthquake, we presented first evidence of this through the empirical variogram of spatially aggregated damage. This motivated the more formal development of a spatial ordinal model which can better separate the two sources of spatial correlation under consideration and better adhere to the Gaussian assumptions commonly made for simulations.

After fitting the proposed model, we illustrated the consequences of this additional spatial correlation on regional portfolio loss estimation: probabilities of extreme loss increased beyond that due to spatial correlated ground motion intensity. By comparing the effects on different building types, the extent of this increase was seen to be linked to the relative sizes of the spatially correlated error and the random error. The smaller the spatially correlated error, the smaller the effect on the regional loss because the random variation in damage outweighs the spatial pattern.

To fit the model to the Haiti dataset, we focused on two of the most common building types (buildings with unreinforced block walls and stone masonry buildings). Theoretically, the model can be extended to jointly model more building categories. However, in practice, we are limited by the high computational burden involving the non-sparse spatial covariance matrices of the Gaussian spatial fields. For our analysis, we used a subset of the available data to complete model optimization in a reasonable amount of time (about six hours). Future work could explore how to speed up computations, incorporate a Bayesian approach and include uncertainty in the model parameter estimates for simulations. It would also be interesting to see how the damage spatial correlation range varies according to context and adapt the model for the case where we have sensor information for IM.

REFERENCES

Abrahamson N & Silva W. 2008. *Summary of the Abrahamson & Silva NGA ground-motion relations.* Earthquake Spectra 24(1): 67–97.

Applied Technology Council 2012. Seismic performance assessment of buildings: Volume 1—Methodology. Redwood City: Applied Technological Council.

Argyris N., Ferretti V., French S., Guikema S., Montibeller G. *Advances in Spatial Risk Analysis.* Risk Analysis 39(1):1–8.

Boore D. & Atkinson G. 2008. *Ground-motion prediction equations for the average horizontal component of PGA, PGV, and 5%-damped PSA at spectral periods between 0.01 s and 10.0 s.* Earthquake Spectra 24(1): 99–138.

Chilès J. & Delfiner P. (2nd ed.) 1999. *Geostatistics: Modeling Spatial Uncertainty*. Hoboken: John Wiley & Sons.

Du A., Padgett J.E. & Shafieezadeh A. 2020. *Influence of intensity measure selection on simulation-based regional seismic risk assessment*. Earthquake Spectra 36(2): 647–672.

Goda K., Ren J. 2010. *Assessment of seismic loss dependence using copula.* Risk Analysis 30(7): 1076–1091.

Graf W.P. & Seligson H.A. 2011. *Earthquake damage to wood-framed buildings in the shakeout scenario.* Earthquake Spectra 27(2): 351–373.

Jayaram N. & Baker J.W. 2009. *Correlation model for spatially distributed ground-motion intensities.* Earthquake Engineering & Structural Dynamics 38(15): 1687–1708.

Lallemant D., Kiremidjian A. & Burton H. 2015. *Statistical procedures for developing earthquake damage fragility curves.* Earthquake Engineering & Structural Dynamics 44(9): 1373–1389.

Nguyen M. & Lallemant D. 2021. *Order matters: Demonstrating the Benefits of Ordinal Fragility Curves for Damage Estimation*. Risk Analysis. https://doi.org/10.1111/risa.13815

Pagani M., Monelli D., Weatherill G., et al. 2014. Openquake engine: An open hazard (and risk) software for the global earthquake model. Seismological Research Letters 85(3): 692–702.

Park J., Bazzurrro P. & Baker J.W. 2007. Modeling spatial correlation of ground motion intensity measures for regional seismic hazard and portfolio loss estimation. In J. Kanda, T. Takada, H. Furuta (Eds.), *Applications of Statistics and Probability in Civil Engineering*, London: Taylor & Francis Group, pp. 1-8.

Silva V. 2019. *Uncertainty and correlation in seismic vulnerability functions of building classes.* Earthquake Spectra 35(4):1515–1539.

Shome N., Jayaram N. & Rahnama M. 2012. *Uncertainty and Spatial Correlation Models for Earthquake Losses*. 15th World Conference on Earthquake Engineering (15WCEE) 10.

Wagenaar D. 2012. *The Significance of Flood Duration for Flood Damage Assessment.* PhD thesis. Delft University of Technology.

Weatherill G., Silva V., Crowley H. & Bazzurro P. 2015. *Exploring the impact of spatial correlations and uncertainties for portfolio analysis in probabilistic seismic loss estimation.* Bulletin of Earthquake Engineering 13(4): 957–981.

Zafarani H., Ghafoori S., Adlaparvar M. 2021. *Spatial Correlation of Peak Ground Motions and Pseudo Spectral Acceleration Based on the Iranian Multievent Datasets.* Journal of Earthquake Engineering. https://doi.org/10.1080/13632469.2021.1911882

Zeng D., Zhang H., Wang C. 2020. *Modelling correlated damage of spatially distributed building portfolios under scenario tropical cyclones.* Structural Safety 87:101978.